

Elements of Statistics and Probability

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Outliers

- **Outliers:** An outlier is an extremely high or an extremely low data value when compared with the rest of the data values

Interquartile Range (IQR)

- **Interquartile Range (IQR):** An IQR is the difference between the third quartile (Q_3) and first quartile (Q_1) of a dataset
- IQR is an appropriate measure of dispersion when the data is skewed and contains significant outliers
- It measures the spread of the middle 50% of data ($Q_3 - Q_1$)

Detection of Outliers

- To determine whether a data value can be considered as an outlier:
 - 1 Compute Q_1 and Q_3
 - 2 Find the $IQR = Q_3 - Q_1$
 - 3 Compute $1.5 \times IQR$
 - 4 Compute Lower bound: $Q_1 - 1.5 \times IQR$, and Upper bound: $Q_3 + 1.5 \times IQR$
 - 5 Consider a value X (say). If $X < Q_1 - 1.5 \times IQR$ or $X > Q_3 + 1.5 \times IQR$, then X is considered an outlier

Example

- Consider the following dataset:

5, 6, 12, 13, 15, 18, 22, 50

- Can the value of 50 be considered as an outlier?

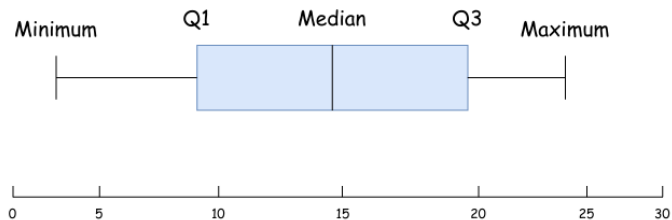
Example (Solution)

- $Q_1 = 9, Q_3 = 20, IQR = 20 - 9 = 11$
- $1.5 \times IQR = 1.5 \times 11 = 16.5$
- Lower bound: $Q_1 - 1.5 \times IQR = 9 - 16.5 = -7.5$
- Upper bound: $Q_3 + 1.5 \times IQR = 20 + 16.5 = 36.5$
- Since the value 50 is outside the range $[-7.5, 36.5]$, 50 is an outlier

Box Plot

- When the data set contains a small number of values, a **box plot** is used to graphically represent the data
- A box plot summarises the data using five key values:
 - Minimum
 - First Quartile (Q_1)
 - Median (Q_2)
 - Third Quartile (Q_3)
 - Maximum

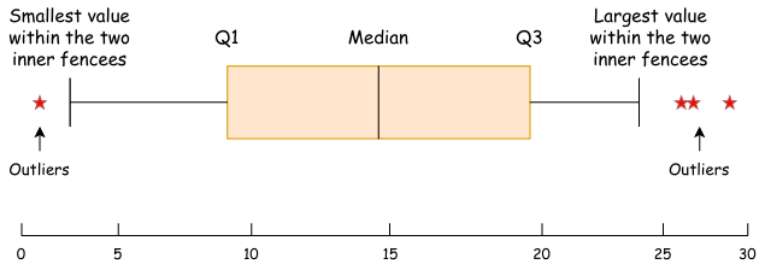
Box Plot



Outlier Detection with Modified Box Plots

- A **modified box plot** is similar to a standard box plot
- Instead of using the minimum and maximum values, it uses:
 - The **smallest** value within the **lower inner fence**
 - The **largest** value within the **upper inner fence**
- Any values outside the inner fences are plotted individually as outliers
- The inner fences are defined as:
 - **Lower inner fence:** $Q_1 - 1.5 \times IQR$
 - **Upper inner fence:** $Q_3 + 1.5 \times IQR$

Modified Box Plot



Example

- The following data are the income (in '000£) for a sample of 12 households:

75, 69, 84, 112, 74, 104, 81, 90, 94, 144, 79, 98

- Construct a box-and-whisker plot for these data and find outliers.

Step 1: Rank and Find Quartiles

- Rank the data in increasing order:

69, 74, 75, 79, 81, 84, 90, 94, 98, 104, 112, 144

- Median = $\frac{84 + 90}{2} = 87$
- $Q_1 = \frac{75 + 79}{2} = 77$
- $Q_3 = \frac{98 + 104}{2} = 101$
- $IQR = Q_3 - Q_1 = 101 - 77 = 24$

Step 2: Compute Inner Fences

- $1.5 \times IQR = 1.5 \times 24 = 36$
- Lower inner fence: $Q_1 - 1.5 \times IQR = 77 - 36$
- Upper inner fence: $Q_3 + 1.5 \times IQR = 101 + 36 = 137$

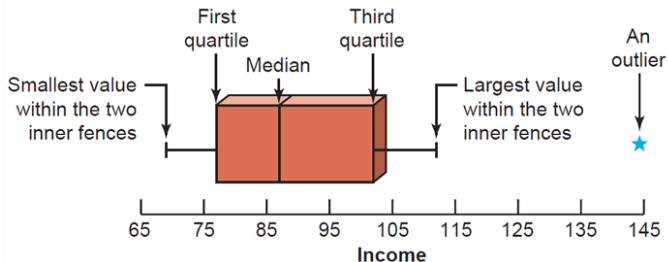
Step 3: Whisker Limits

- Smallest value within the inner fences: **69**
- Largest value within the inner fences: **112**
- Values outside the fences (possible outliers): **144**

Step 4: Construct the Box-and-Whisker Plot

- Draw a horizontal line covering the range of the data
- Mark the five-number summary:
 - Minimum (within fences): 69
 - Q_1 : 77
 - Median: 87
 - Q_3 : 101
 - Maximum (within fences): 112
- Values outside the inner fences (e.g., 144) are plotted as outliers

Modified Box Plot



Skewness from a Box Plot

Symmetric (Normal Distribution)

The median line is in the **exact center** of the box, and both whiskers are of **approximately equal length**. Outliers, if any, are evenly distributed on both sides.

Right-Skewed (Positive Skew)

The median line is **closer to the bottom** (Q1) of the box, the upper section of the box is longer than the lower, and the **upper whisker is longer** than the lower whisker. This indicates a longer "tail" on the right side of the distribution.

Left-Skewed (Negative Skew)

The median line is **closer to the top** (Q3) of the box, the lower section of the box is longer than the upper, and the **lower whisker is longer** than the upper whisker. This indicates a longer "tail" on the left side of the distribution.

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Stem and Leaf Plot

- A **stem and leaf plot** is a data plot that uses part of a data value as the “stem” (leading digits) and part of the data value as the “leaf” (last digit) to form groups or classes.
- At an outpatient testing center, a sample of 20 days showed the following number of cardiograms done each day: 25, 31, 20, 32, 13, 14, 43, 02, 57, 23, 36, 32, 33, 32, 44, 32, 52, 44, 51, 45.
Construct a stem and leaf plot for the data.

Stem and Plot

Leading Digit (Stem)	Trailing Digit (Leaf)
0	2
1	3 4
2	0 3 5
3	1 2 2 2 2 3 6
4	3 4 4 5
5	1 2 7

Key: $3|2 = 32$

Thank You